

JASWANTH KUMAR N

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EDUCATION

Amrita Vishwa Vidyapeetham, Bengaluru	2022 – 2026
<i>Bachelor of Technology (B.Tech) - Electronics and Computer Engineering</i>	<i>CGPA: 7.38 / 10.0</i>
Sri Chaitanya Techno School, Bengaluru	2022
<i>Class XII - CBSE</i>	86%
St. Francis School, Bengaluru	2020
<i>Class X - ICSE</i>	92%

EXPERIENCE

Backend Developer Intern	Jan 2026 – Sep 2026
<i>AES Technologies India Private Limited</i>	
– Developed and tested RESTful APIs using Flask and MySQL, working with JSON request/response data, HTTP methods, status codes, query parameters, request validation, and authentication. – Designed modular backend components with routes, controllers, models, and configuration modules, implementing request validation, secure authentication, and exception handling. – Tested and debugged API endpoints using Postman Collections, validating request and response behavior and documenting API functionality for reliable backend development. – Validated database operations through phpMyAdmin and handled HTTP responses to support accurate API functionality and troubleshooting.	

PROJECTS

Truck Platooning & Route Optimization (Publication)	
<i>Python, Google Maps API, Google OR-Tools</i>	
– Integrated Google Maps API to retrieve distance and duration data, generate distance matrices, and support route optimization across multiple delivery locations. – Developed a vehicle routing optimization system using Python and Google OR-Tools to solve Capacitated Vehicle Routing Problem (CVRP) scenarios with capacity constraints and platoon formation. – Evaluated routing performance across different scenarios using distance, travel-duration, vehicle-utilization, and cost-based comparisons to identify efficient delivery plans and improve fleet utilization.	
Adaptive Evolving Online Firewall (Publication)	
<i>Python, Flask, AWS, Isolation Forest, ADWIN</i>	
– Developed an adaptive cloud firewall using Python and Flask for real-time anomaly detection in evolving network traffic. – Implemented streaming anomaly detection with Online Isolation Forest, per-source behavioral tracking, and ADWIN-based concept drift adaptation. – Deployed and evaluated the framework on AWS using Elastic Beanstalk, CloudFront, and S3, achieving 100% precision, 82.2% F1-score, and 99.2% ROC-AUC on benchmark datasets.	
Sugarcane Crop Phenology Analysis	
<i>Python, Google Earth Engine, XGBoost, LSTM</i>	
– Developed an end-to-end remote sensing pipeline using Sentinel-2 NDVI time-series imagery and meteorological data from 2020–2024 to analyze sugarcane crop phenology across multiple districts in Tamil Nadu. – Processed and validated agricultural datasets, engineered lag and rolling-window features, and trained Random Forest and XGBoost models to classify five phenological growth stages, achieving 83.37% classification accuracy with Random Forest.	

SKILLS

Programming: Python, SQL
API & Backend: REST APIs, Flask, Postman, JSON, HTTP Methods, HTTP Status Codes, Query Parameters, Request Validation, Authentication
Data Analysis: Pandas, NumPy, Microsoft Excel, Google Sheets, Spreadsheets, Data Validation, Data Quality Analysis, EDA
Tools: MySQL, Git, GitHub, AWS, Google Maps API, Google Earth Engine

CERTIFICATIONS

AWS Academy: Cloud Foundations, Cloud Operations, Machine Learning Foundations

LEADERSHIP

Vice President	Aug 2024 – Jul 2025
<i>Smriti – Photography & Design Club, Amrita Vishwa Vidyapeetham</i>	
– Coordinated photography events, workshops, competitions, and photowalks while managing club operations and student volunteers.	